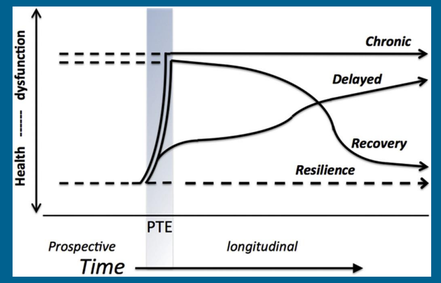
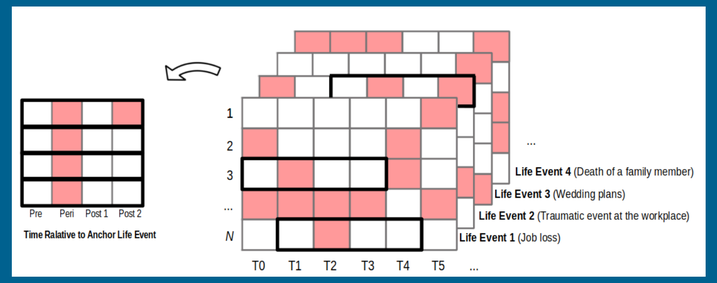
Mental Health Trajectories in Response to Adversity

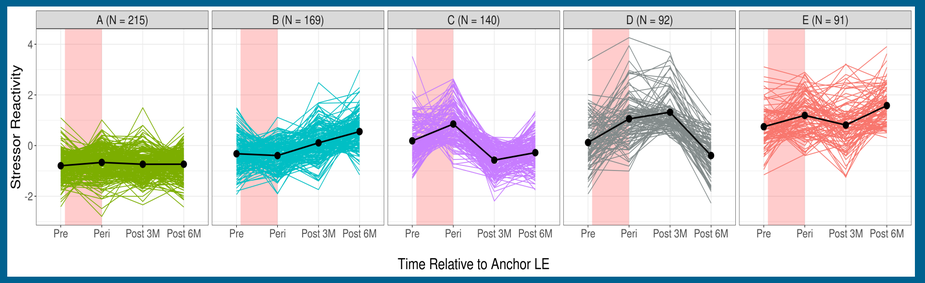
A Longitudinal Clustering Approach

Charlotte Schenk, M.Sc. Psychology; c.schenk@med.uni-frankfurt.de
Dept. of Psychiatry, Psychosomatic Medicine and Psychotherapy, University Medical Centre Frankfurt

LORA RESILIENZ STUDIE



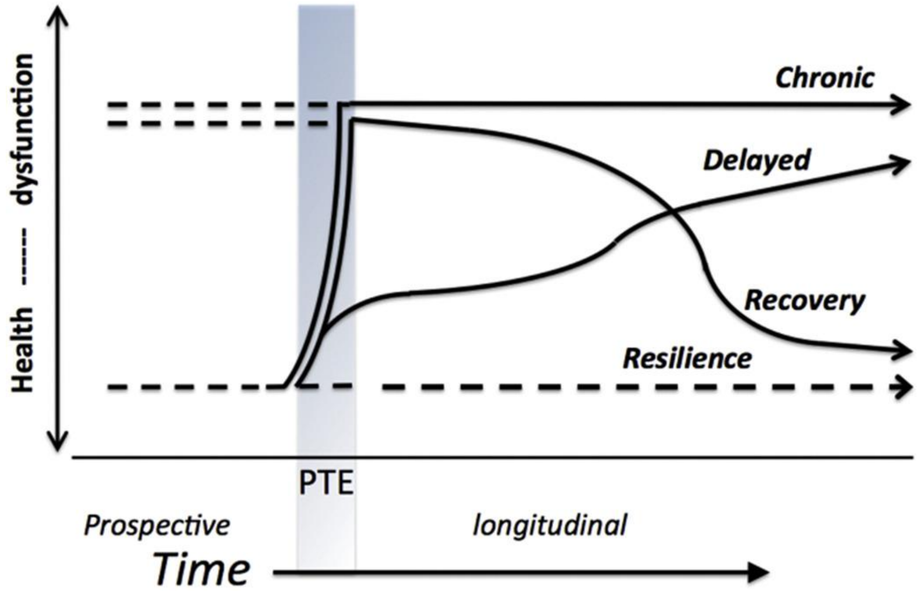




Resilience

Resilience: Maintenance or quick recovery of mental health during or after stressor exposure.[1]

Figure 1
Commonly observed longitudinal trajectories of response to potential trauma.

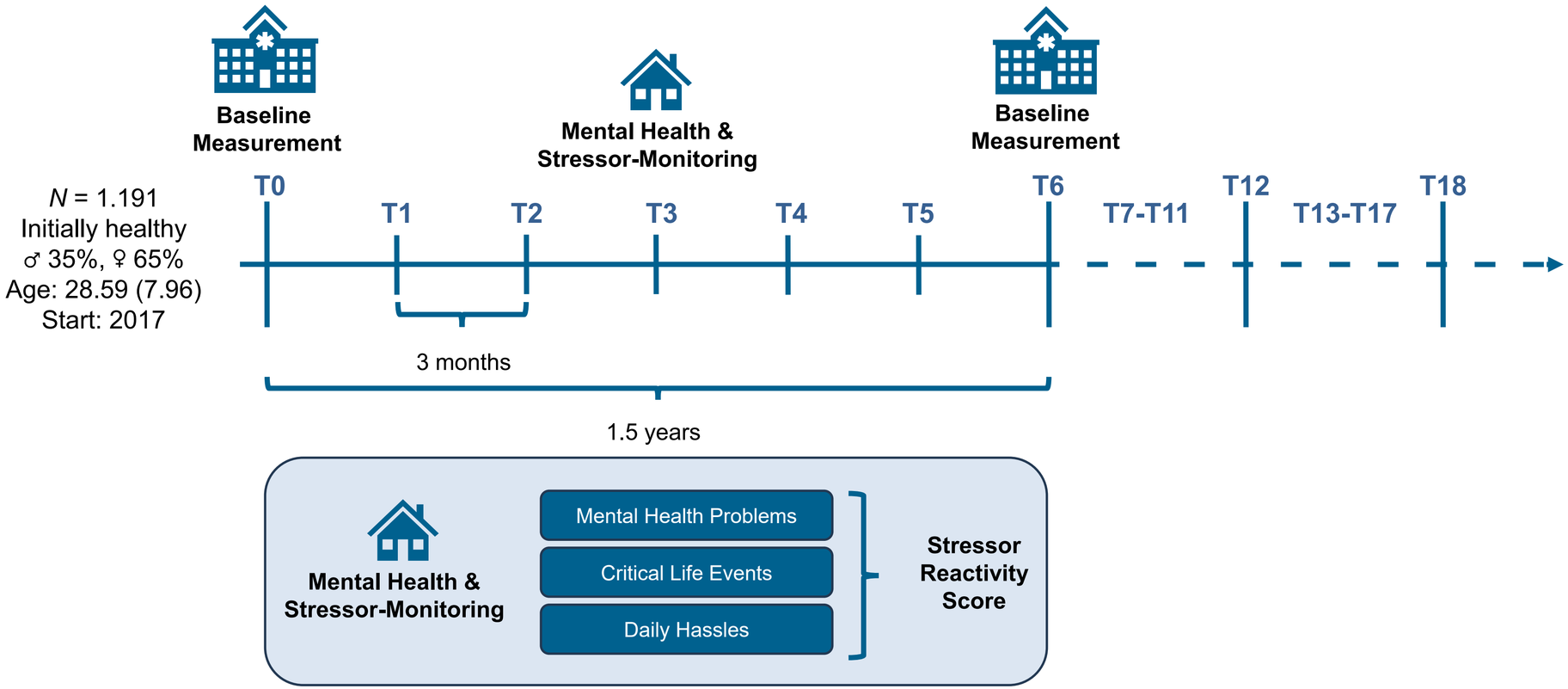


Note. Sourced from Galatzer-Levy et al. (2018).[2]

[1] Kalisch, R., Müller, M. B., & Tüscher, O. (2015). A conceptual framework for the neurobiological study of resilience. *Behavioral and brain sciences,* 38, e92.

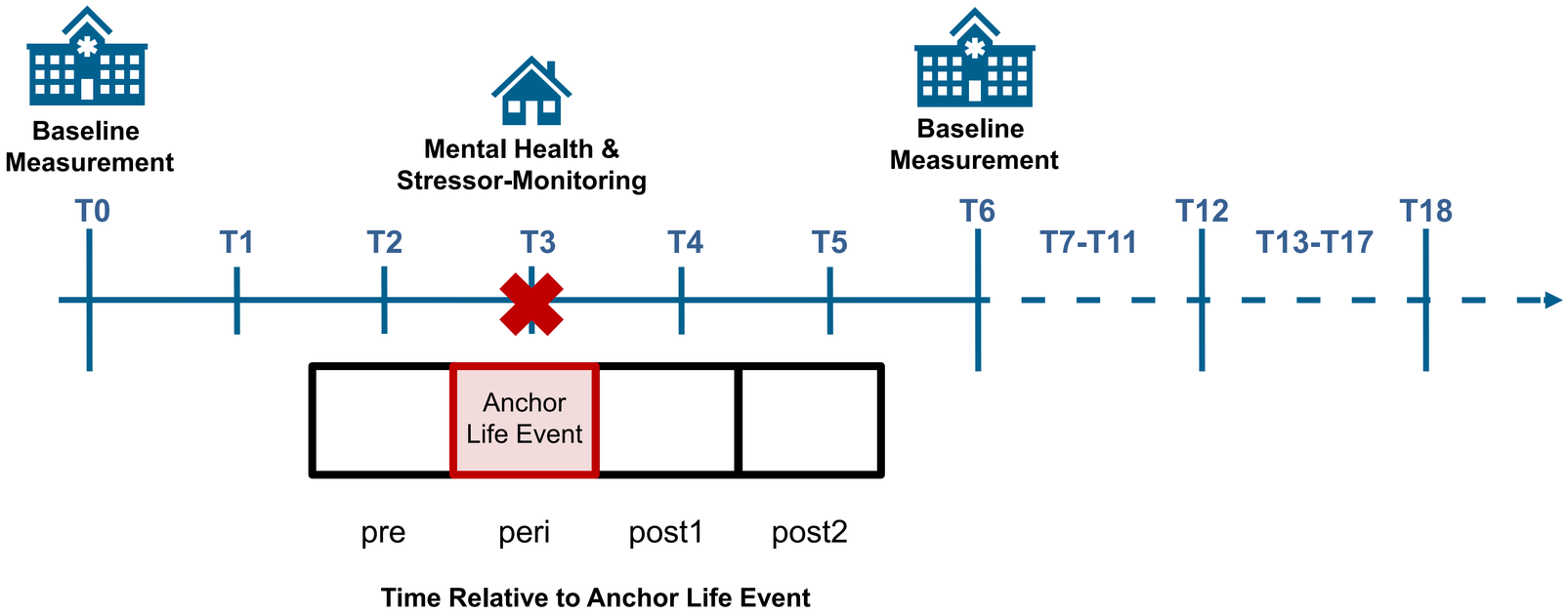
[2] Galatzer-Levy, I. R., Huang, S. H., & Bonanno, G. A. (2018). Trajectories of resilience and dysfunction following potential trauma: A review and statistical evaluation. *Clinical psychology review*, 63, 41-55.

LORA - LOngitudinal Resilience Assessment[1]



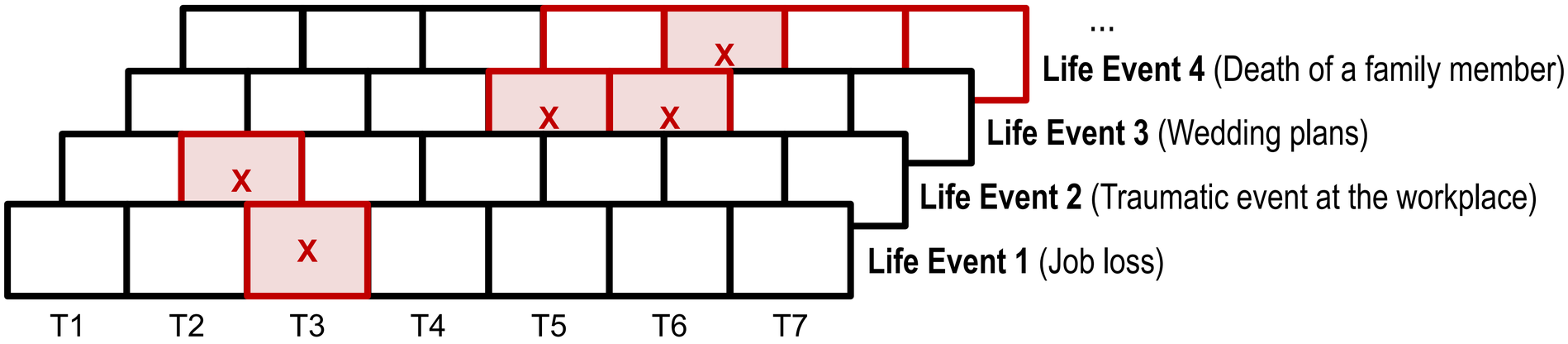
[1] Chmitorz, A., Neumann, R. J., Kollmann, B., Ahrens, K. F., Öhlschläger, S., Goldbach, N., ... & Reif, A. (2021). Longitudinal determination of resilience in humans to identify mechanisms of resilience to modern-life stressors: the longitudinal resilience assessment (LORA) study. *European archives of psychiatry and clinical neuroscience*, 271, 1035-1051.

Event Centered Time Frame[1]



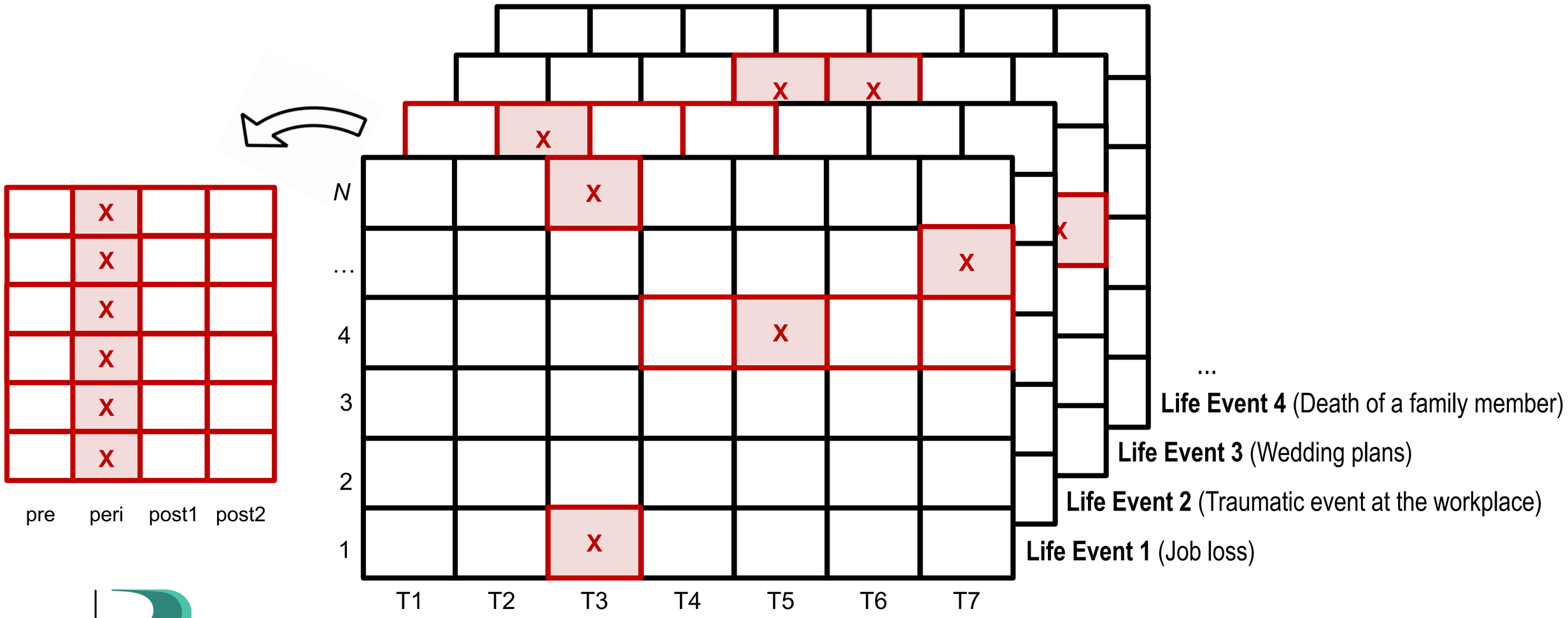
[1] Ahrens, K. F., Schenk, C., Kollmann, B., Puhlmann, L., Neumann, R. J., Schäfer, S. K., ... & Plichta, M. M. (2024). Resilience to major life events: Advancing trajectory modeling and resilience factor identification by controlling for background stressor exposure. *American Psychologist*, *79*(8), 1076.

Event Centered Time Frame[1]



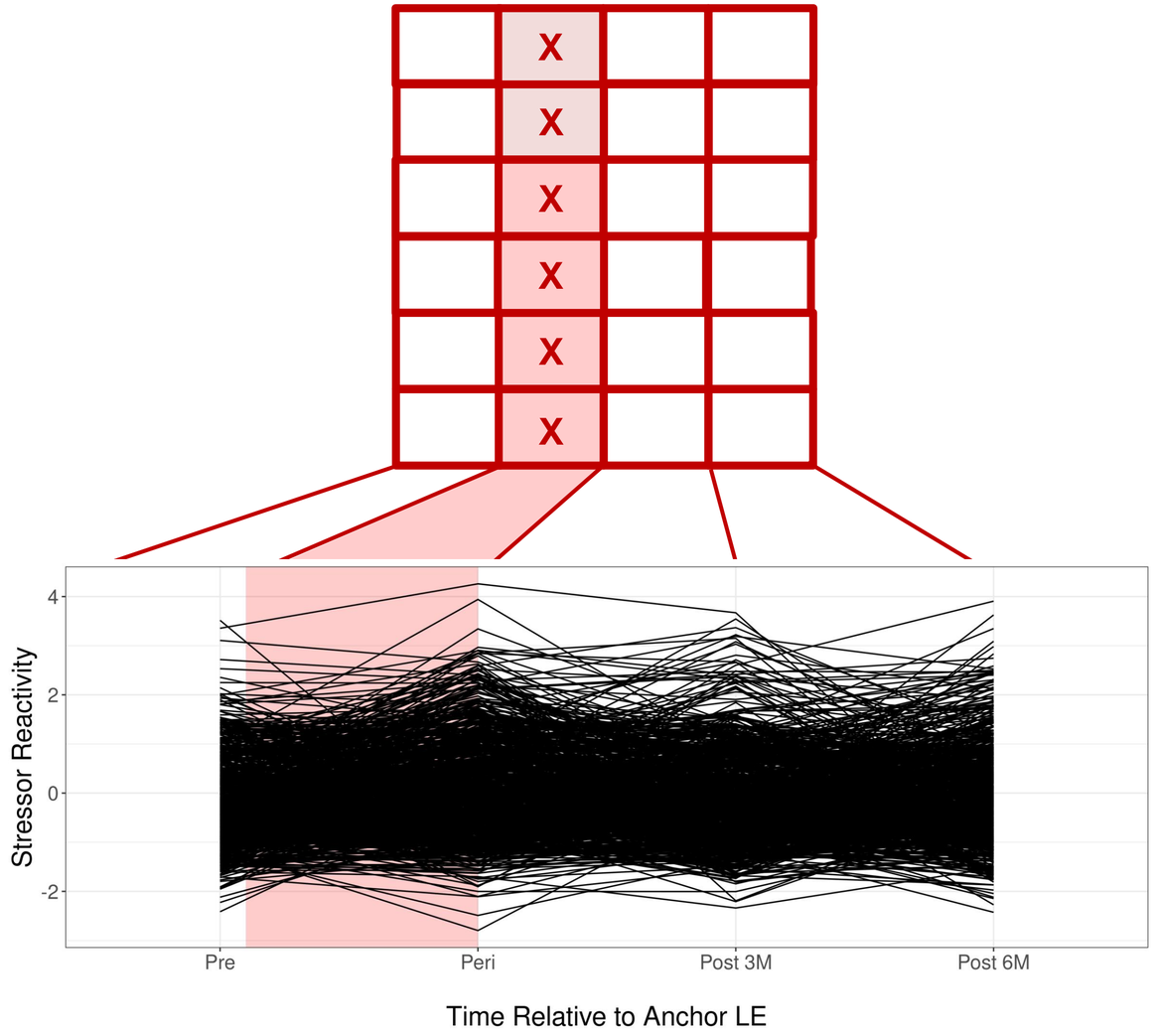
[1] Ahrens, K. F., Schenk, C., Kollmann, B., Puhlmann, L., Neumann, R. J., Schäfer, S. K., ... & Plichta, M. M. (2024). Resilience to major life events: Advancing trajectory modeling and resilience factor identification by controlling for background stressor exposure. *American Psychologist*, *79*(8), 1076.

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Selected Methods for Clustering Longitudinal Data[1,2]

Longitudinal k-means

Description

- Unsupervised, data-driven method using an iterative assignment–update procedure
- Groups individuals with similar trajectories based on a **distance metric**
- Each cluster is represented by a **mean trajectory (centroid)**

Advantages

- Can capture **complex/non-linear trajectory shapes**
- **No distributional assumptions**
- **Computationally efficient**

Limitations

- Typically requires **complete, balanced, and aligned data**
- Time is **not explicitly modeled**
- Covariates cannot be **directly incorporated into the clustering procedure**

[1] Lu, Z. (2025). Clustering longitudinal data: a review of methods and software packages. *International Statistical Review*, *93*(3), 425-458.
[2] Den Teuling, N., Pauws, S., & van den Heuvel, E. (2021). Clustering of longitudinal data: A tutorial on a variety of approaches. *arXiv e-prints*, arXiv-2111.

Selected Methods for Clustering Longitudinal Data[1,2]

	Longitudinal k-means	LCGA
Description	• Unsupervised, data-driven method using an iterative assignment–update procedure • Groups individuals with similar trajectories based on a **distance metric** • Each cluster is represented by a **mean trajectory (centroid)**	• Model-based approach assuming a number of latent trajectory classes • Each class is characterized by its own **growth trajectory** (e.g., intercept and slope) • Individuals are assigned based on their **posterior probability of class membership**
Advantages	• Can capture **complex/non-linear trajectory shapes** • **No distributional assumptions** • **Computationally efficient**	• **Time is explicitly modeled** • Can include **covariates** • Relatively computationally efficient • Can model non-linear trajectories
Limitations	• Typically requires **complete, balanced, and aligned data** • Time is **not explicitly modeled** • Covariates cannot be **directly incorporated into the clustering procedure**	• Assumes **conditional independence** of repeated measurements given class membership • Assumes little/no individual variation around the class-specific growth trajectory • Requires **distributional/model assumptions**

[1] Lu, Z. (2025). Clustering longitudinal data: a review of methods and software packages. *International Statistical Review*, *93*(3), 425-458.
[2] Den Teuling, N., Pauws, S., & van den Heuvel, E. (2021). Clustering of longitudinal data: A tutorial on a variety of approaches. *arXiv e-prints*, arXiv-2111.

Selected Methods for Clustering Longitudinal Data[1,2]

	Longitudinal k-means	LCGA	LGMM
Description	• Unsupervised, data-driven method using an iterative assignment–update procedure • Groups individuals with similar trajectories based on a **distance metric** • Each cluster is represented by a **mean trajectory (centroid)**	• Model-based approach assuming a number of latent trajectory classes • Each class is characterized by its own **growth trajectory** (e.g., intercept and slope) • Individuals are assigned based on their **posterior probability of class membership**	• Same basic framework as LCGA, but allows **within-class heterogeneity through random effects** • Individuals can **deviate from the class-specific trajectory** • Individuals are assigned based on their **posterior probability of class membership**
Advantages	• Can capture **complex/non-linear trajectory shapes** • **No distributional assumptions** • **Computationally efficient**	• **Time is explicitly modeled** • Can include **covariates** • Relatively computationally efficient • Can model non-linear trajectories	• **Time is explicitly modeled** • Can include **covariates** • Allows **within-class variation through random effects** • Accounts for **within-person correlation** between repeated measurements
Limitations	• Typically requires **complete, balanced, and aligned data** • Time is **not explicitly modeled** • Covariates cannot be **directly incorporated into the clustering procedure**	• Assumes **conditional independence** of repeated measurements given class membership • Assumes little/no individual variation around the class-specific growth trajectory • Requires **distributional/model assumptions**	• **Computationally intensive** • More prone to **convergence issues** • Requires additional **distributional assumptions**, including for random effects

[1] Lu, Z. (2025). Clustering longitudinal data: a review of methods and software packages. *International Statistical Review*, *93*(3), 425-458.
[2] Den Teuling, N., Pauws, S., & van den Heuvel, E. (2021). Clustering of longitudinal data: A tutorial on a variety of approaches. *arXiv e-prints*, arXiv-2111.

Trajectories - Clustering by k-means for longitudinal data



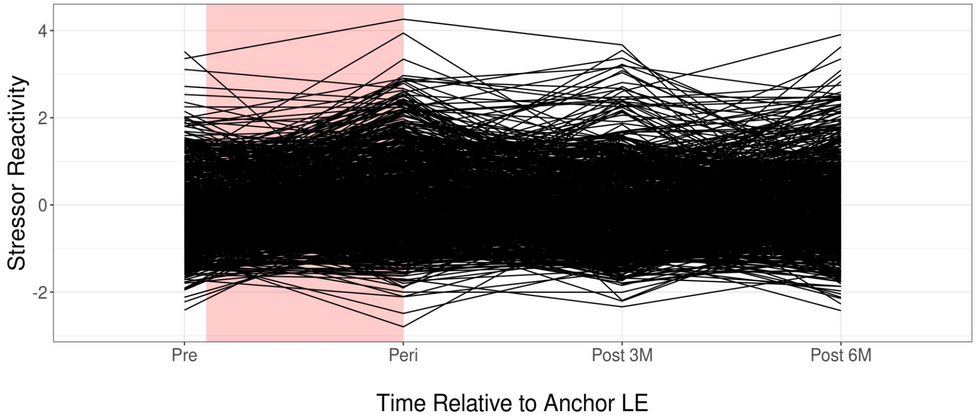
Dr. Kira Ahrens

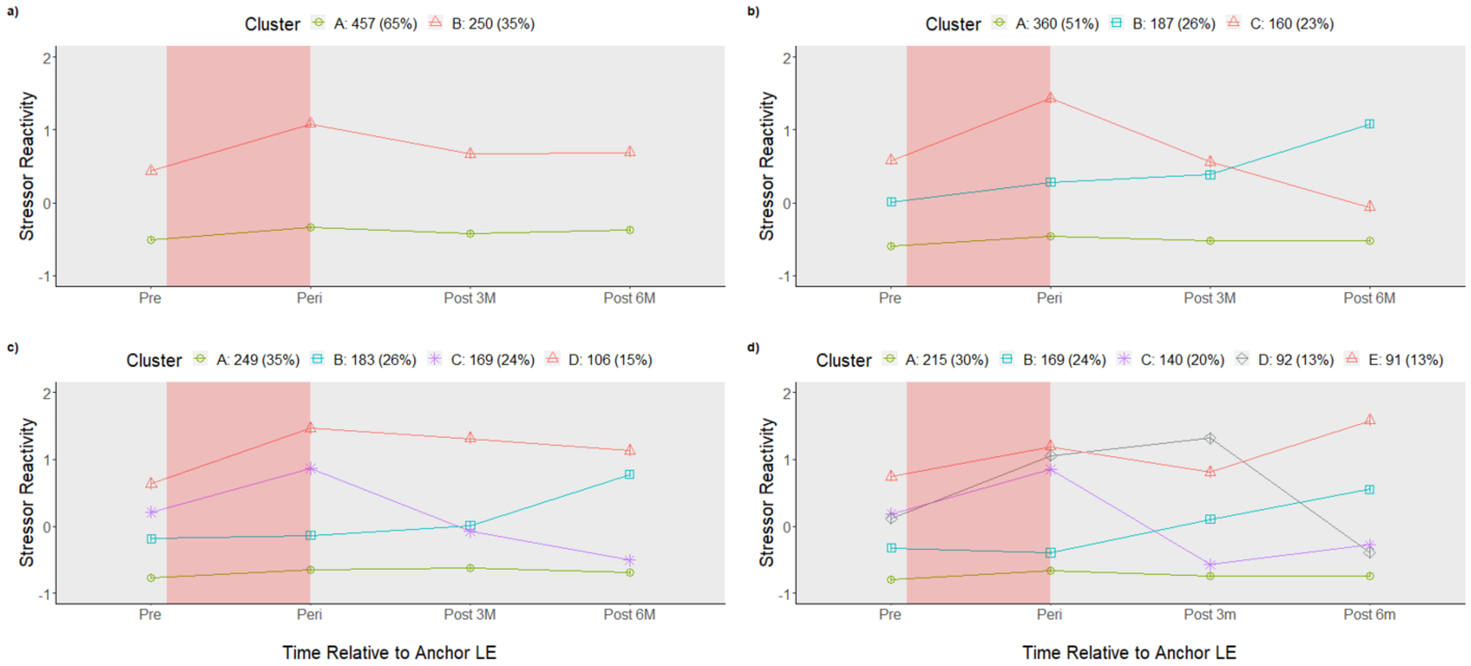


Prof. Dr. Raffael Kalisch



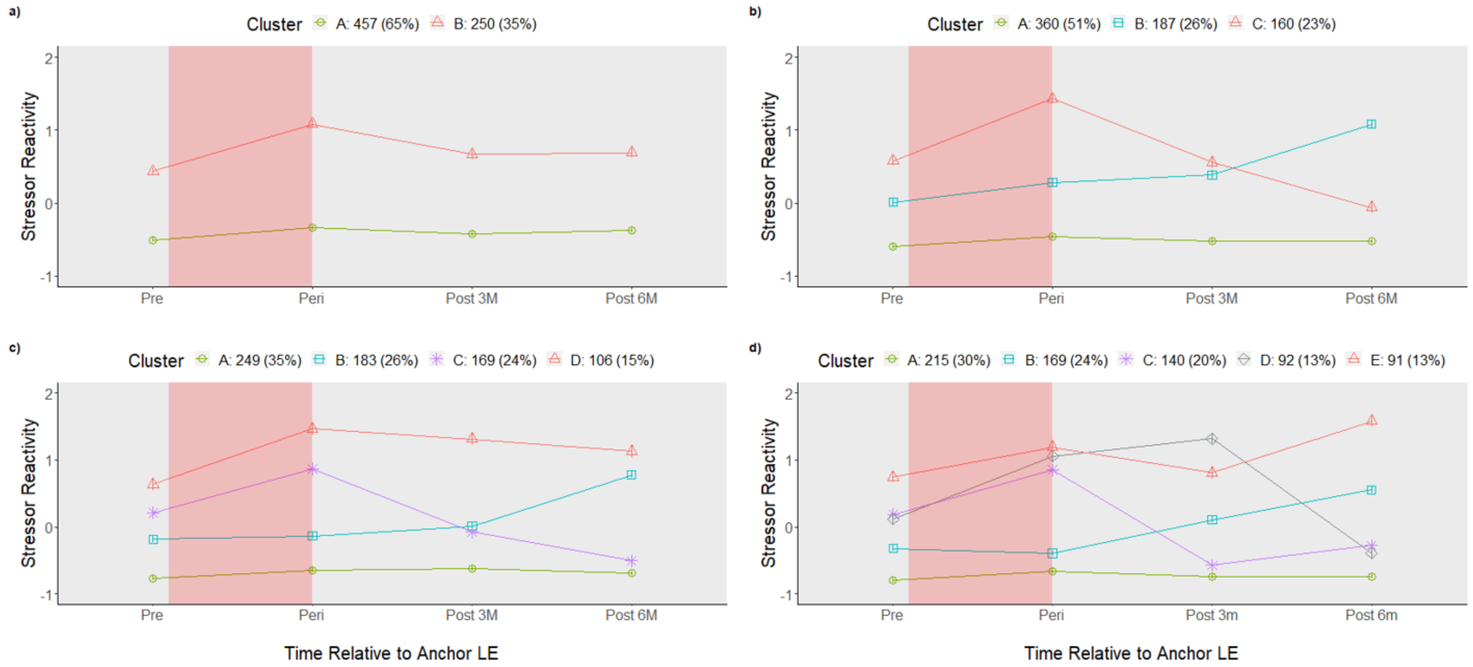
Prof. Dr. Michael M. Plichta

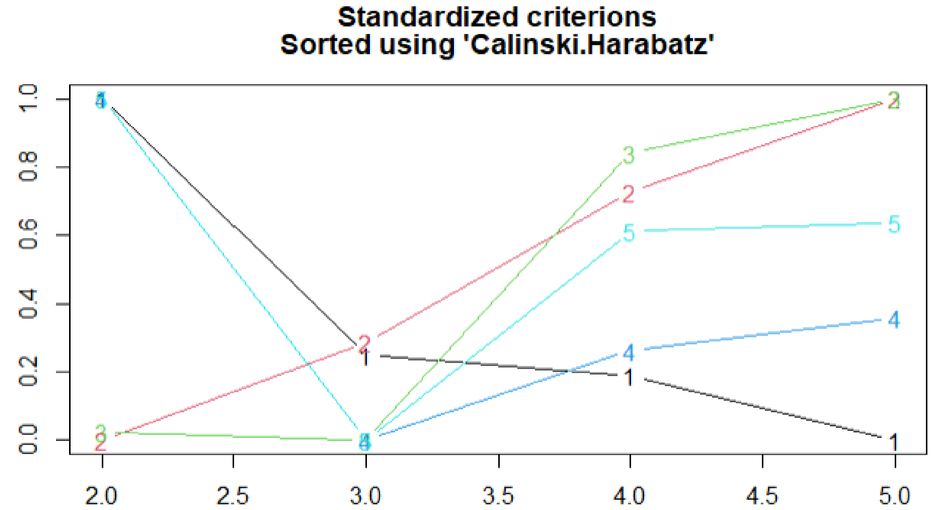




[1] Ahrens, K. F., Schenk, C., Kollmann, B., Puhlmann, L., Neumann, R. J., Schäfer, S. K., ... & Plichta, M. M. (2024). Resilience to major life events: Advancing trajectory modeling and resilience factor identification by controlling for background stressor exposure. *American Psychologist*, *79*(8), 1076.

k-means Clustering for longitudinal Data

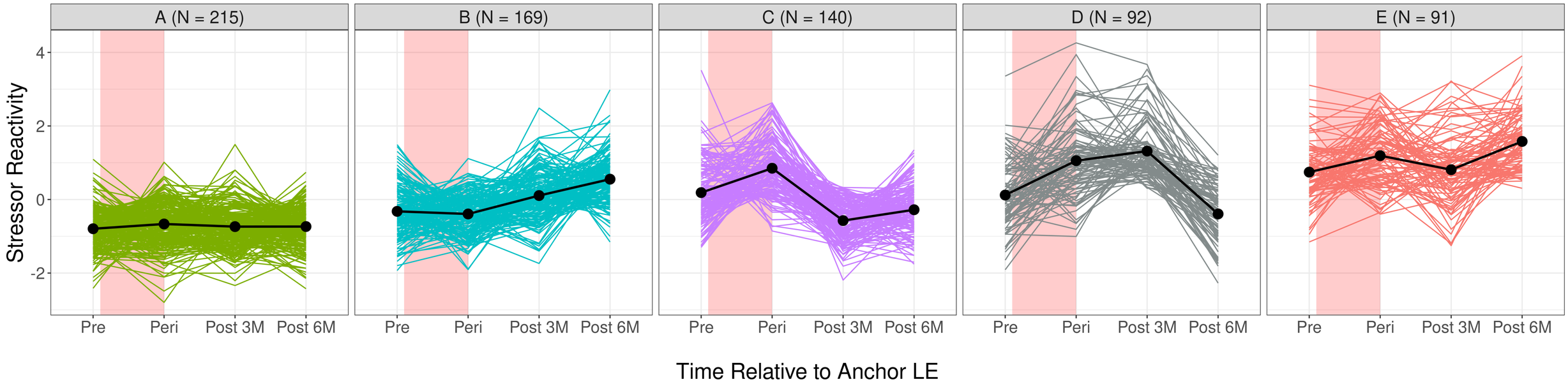




IMPORTANT NOTE: Some criterion should be maximized, some other should be minimized. This might be confusing for the non expert. In order to simplify the comparison of the criterion, `qualityCriterion` compute the OPPOSITE of the criterion that should be minimized (Ray & Bouldin, Davies & Turi, BIC and AIC). Thus, all the criterion computed by this function should be maximized.

https://cran.r-project.org/web/packages/longitudinalData/longitudinalData.pdf

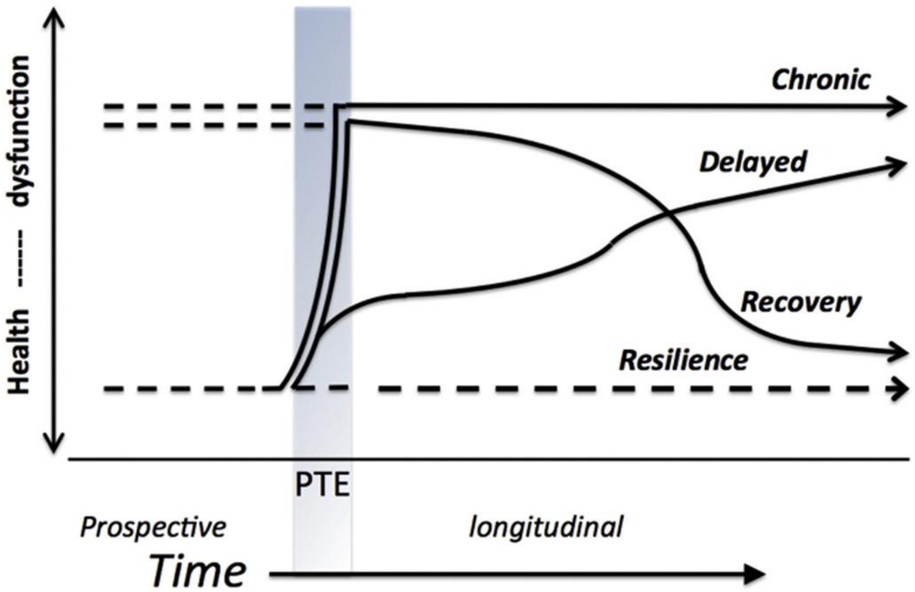
Trajectories - Clustering by k-means for longitudinal data



[1] Ahrens, K.F., Schenk, C., Kollmann, B.,… , Reif, A., Kalisch, R, & Plichta, M. M. (in press). Resilience to Major Life Events: Advancing Trajectory Modeling and Resilience Factor Identification by Controlling for Background Stressor Exposure. *American Psychologist.* https://doi.org/10.1037/amp0001315

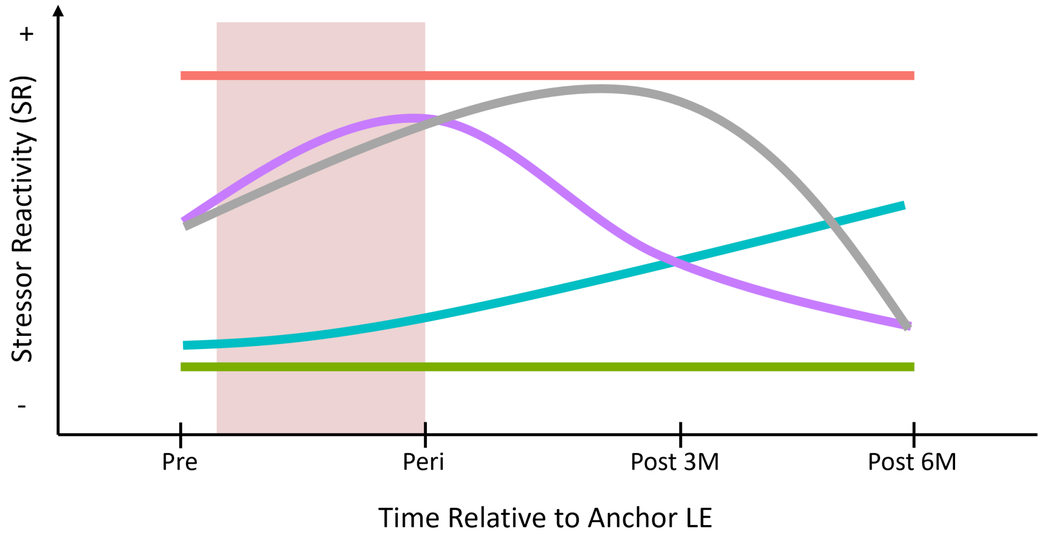
Trajectories - Clustering by k-means for longitudinal data

Commonly observed longitudinal trajectories of response to potential trauma.



Note. Sourced from Galatzer-Levy et al. (2018).[1]

Schematically illustrated trajectories of the presented study.

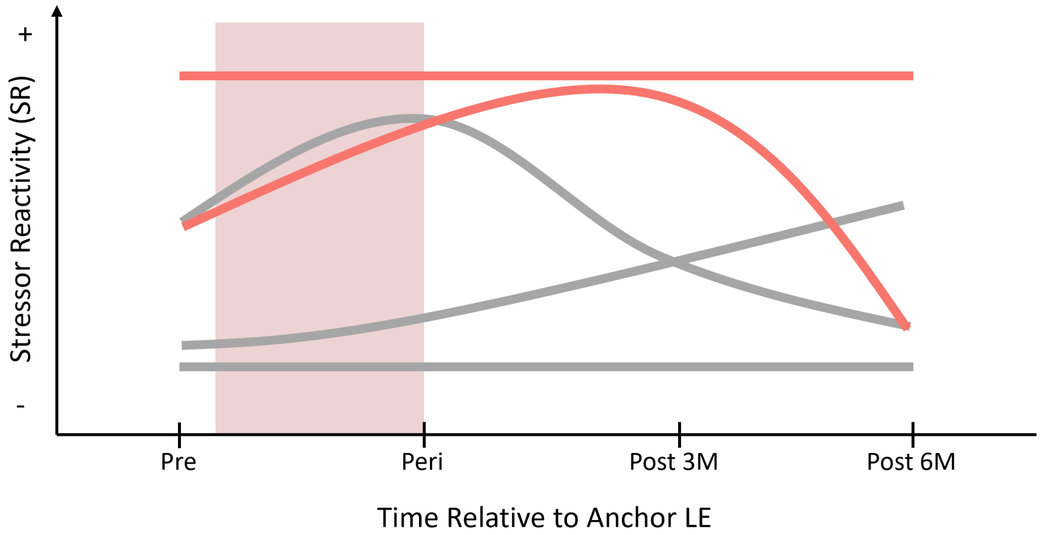


Note. In order to cluster the trajectories, not the raw mental health scores but the stressor reactivity was used to consider the background stressors.

[1] Galatzer-Levy, I. R., Huang, S. H., & Bonanno, G. A. (2018). Trajectories of resilience and dysfunction following potential trauma: A review and statistical evaluation. *Clinical psychology review*, 63, 41-55.

Validity of the Trajectories

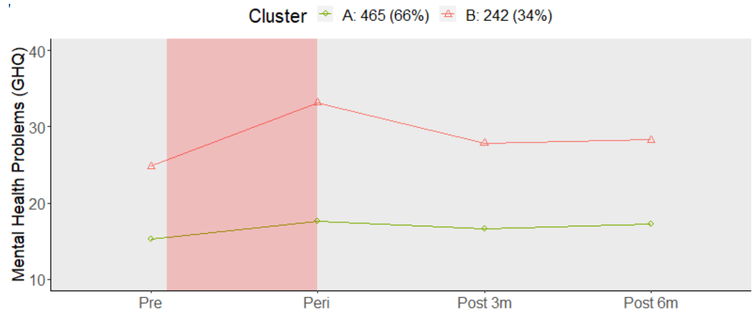
Construct validity:
The resilience factor *sense of coherence* significantly predicts cluster membership.



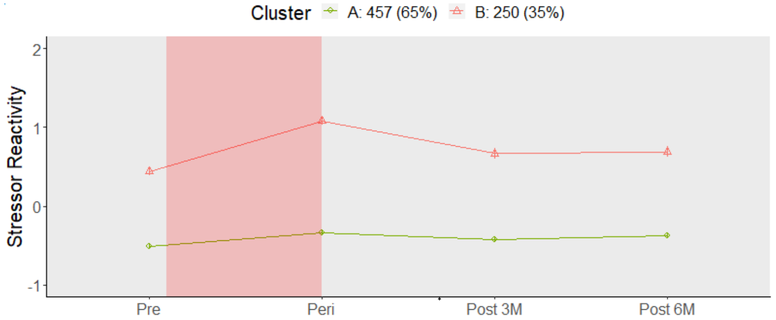
Predictive validity:
Cluster membership significantly predicts mental disorders.

[1] Ahrens, K. F., Schenk, C., Kollmann, B., Puhlmann, L., Neumann, R. J., Schäfer, S. K., ... & Plichta, M. M. (2024). Resilience to major life events: Advancing trajectory modeling and resilience factor identification by controlling for background stressor exposure. *American Psychologist*, *79*(8), 1076.

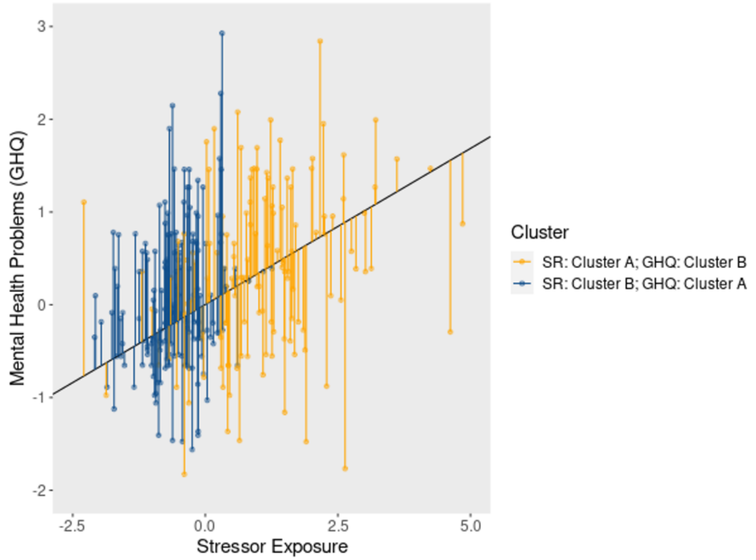
Comparison of Stressor Reactivity vs. Mental Health Trajectory Clustering



Raw Mental Health Scores: *How good or poor is an individual's mental health?*



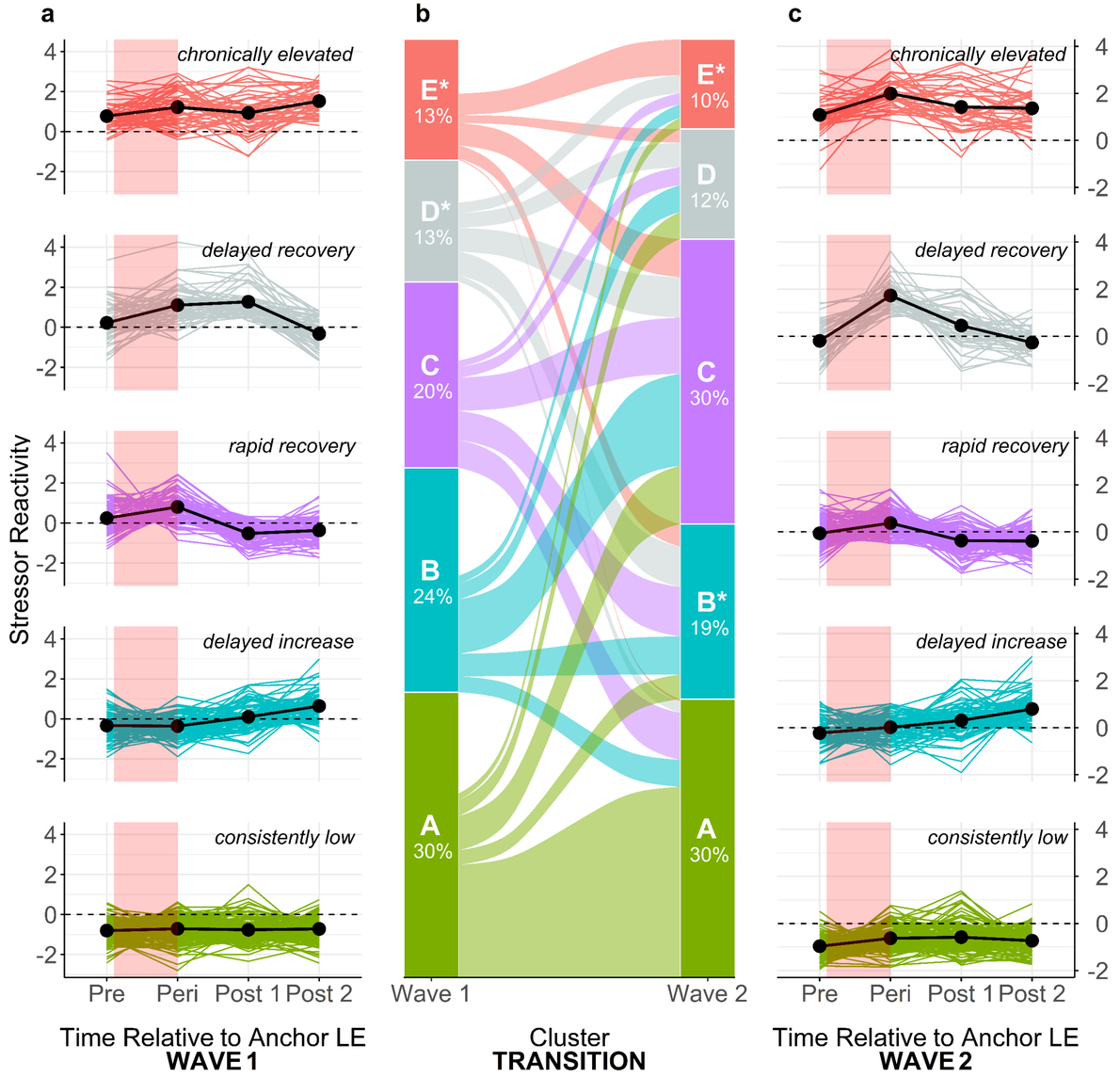
Stressor ReactivityScores: *How good or poor is an individual's mental health relative to experienced stressor exposure?*



The choice of outcome measure changed trajectory classification depending on individual stressor exposure.

[1] Ahrens, K. F., Schenk, C., Kollmann, B., Puhlmann, L., Neumann, R. J., Schäfer, S. K., ... & Plichta, M. M. (2024). Resilience to major life events: Advancing trajectory modeling and resilience factor identification by controlling for background stressor exposure. *American Psychologist*, *79*(8), 1076.

Individual Dynamics of Trajectories Across Successive Life Events



Schenk, C., Ahrens, K.F.,… , Kalisch, R, Reif, A. & Plichta, M. M. (in prep). *Individual Dynamics in Psychological Resilience*

Thank you for your attention! Are there any questions?

k-means Clustering



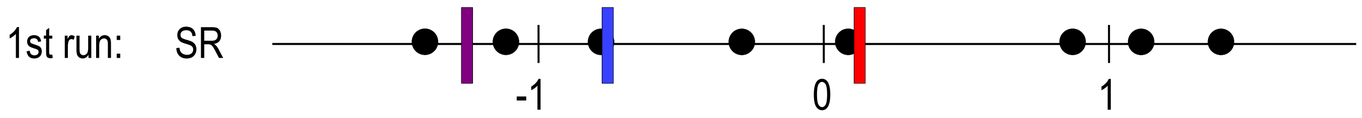
k-means Clustering

1. Choosing the number of clusters: $k = 3$



k-means Clustering

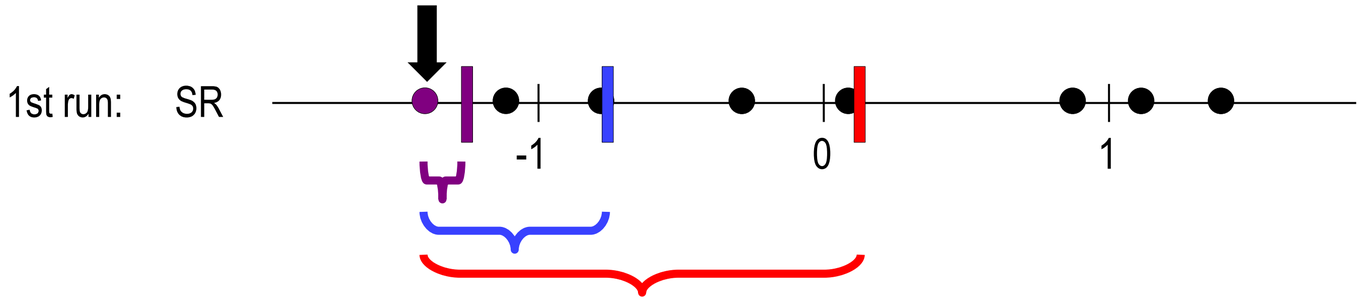
1. Choosing the number of clusters: *k* = 3
2. **Randomly assign cluster centers (centroids)**



1. cluster centroid (purple cluster)
2. cluster centroid (blue cluster)
3. cluster centroid (red cluster)

k-means Clustering

1. Choosing the number of clusters: *k* = 3
2. Randomly assign cluster centers (centroids)
3. **Assign data points to the cluster with the nearest centroid**

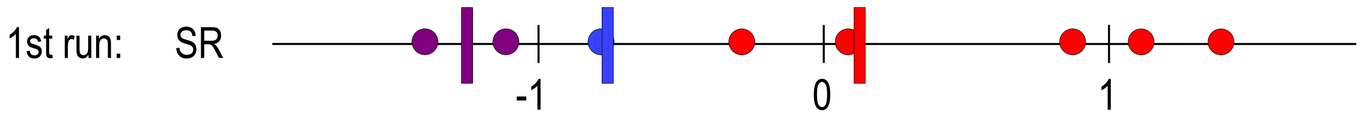


Distances from the first data point to the centroids

1. cluster centroid (purple cluster)
2. cluster centroid (blue cluster)
3. cluster centroid (red cluster)

k-means Clustering

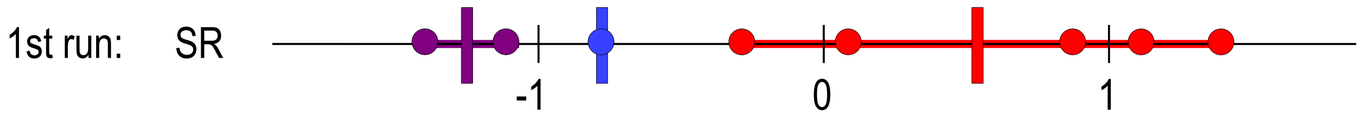
1. Choosing the number of clusters: *k* = 3
2. Randomly assign cluster centers (centroids)
3. **Assign data points to the cluster with the nearest centroid**



1. cluster centroid (purple cluster)
2. cluster centroid (blue cluster)
3. cluster centroid (red cluster)

k-means Clustering

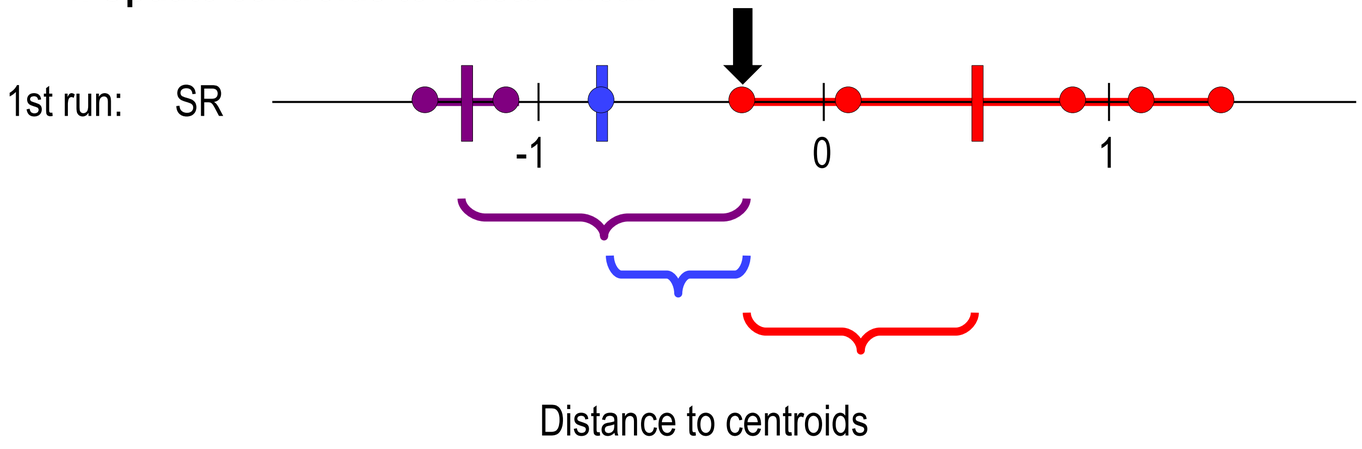
1. Choosing the number of clusters: *k* = 3
2. Randomly assign cluster centers (centroids)
3. Assign data points to the cluster with the nearest centroid
4. **Update centroids to cluster mean**



1. cluster centroid (purple cluster)
2. cluster centroid (blue cluster)
3. cluster centroid (red cluster)

k-means Clustering

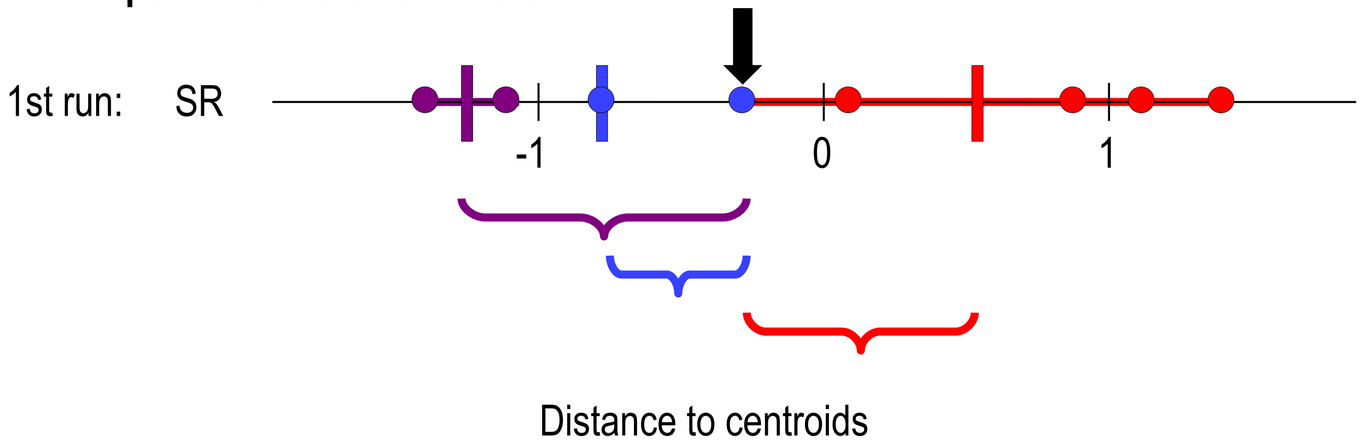
1. Choosing the number of clusters: $k = 3$
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3. **Assign data points to the cluster with the nearest centroid**
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1. cluster centroid (purple cluster)
2. cluster centroid (blue cluster)
3. cluster centroid (red cluster)

k-means Clustering

1. Choosing the number of clusters: $k = 3$
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3. **Assign data points to the cluster with the nearest centroid**
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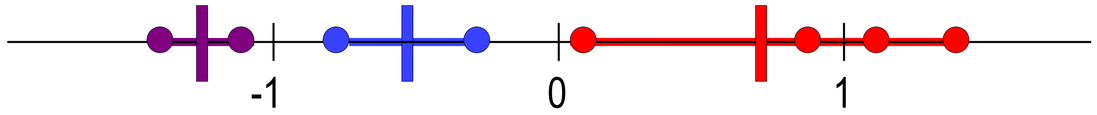


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2. cluster centroid (blue cluster)
3. cluster centroid (red cluster)

k-means Clustering

1. Choosing the number of clusters: *k* = 3
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3. **Assign data points to the cluster with the nearest centroid**
4. **Update centroids to cluster mean**

1st run: SR

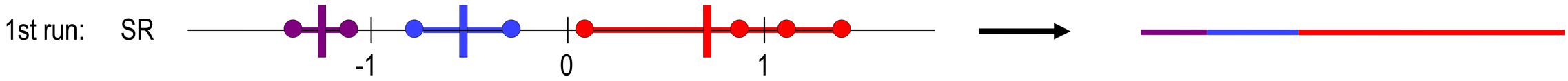


1. cluster centroid (purple cluster)
2. cluster centroid (blue cluster)
3. cluster centroid (red cluster)

k-means Clustering

1. Choosing the number of clusters: $k = 3$
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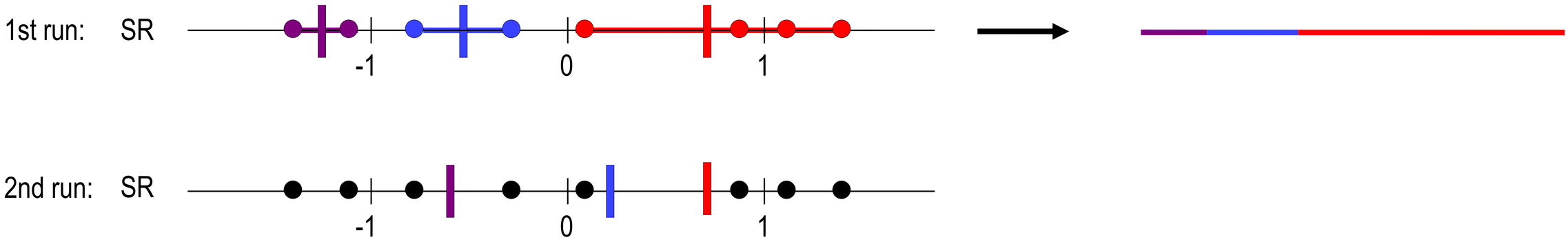
Total within-cluster variance:



k-means Clustering

1. Choosing the number of clusters: $k = 3$
2. **Randomly assign cluster centers (centroids)**
3. Assign data points to the cluster with the nearest centroid
4. Update centroids to cluster mean

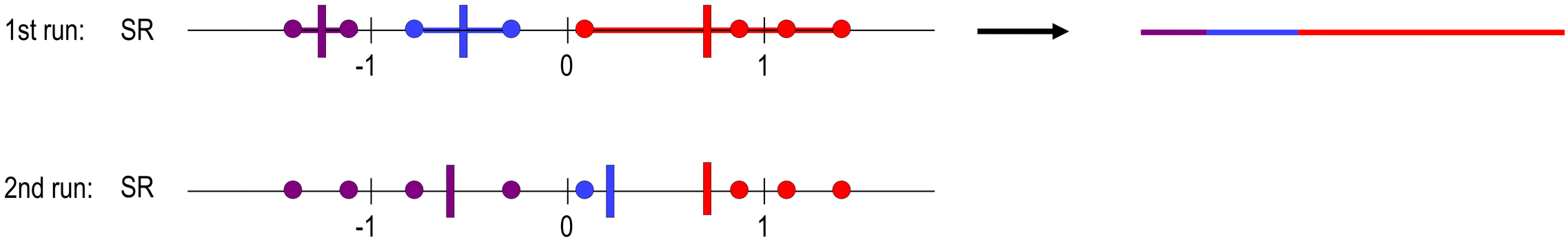
Total within-cluster variance:



k-means Clustering

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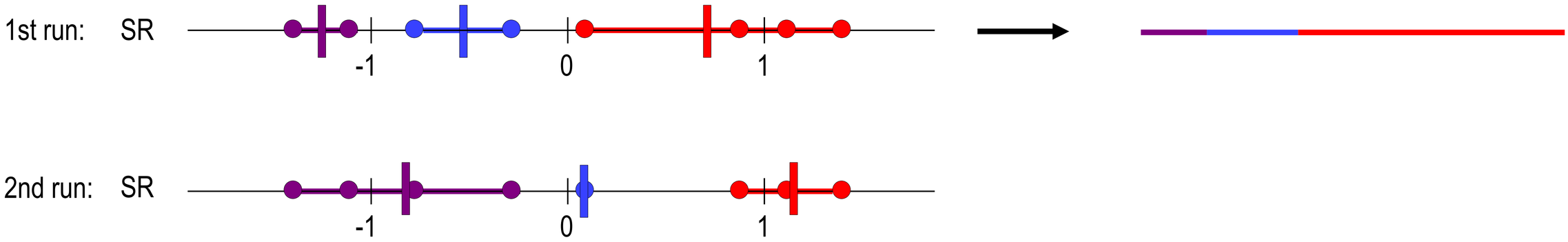
Total within-cluster variance:



k-means Clustering

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2. Randomly assign cluster centers (centroids)
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4. **Update centroids to cluster mean**

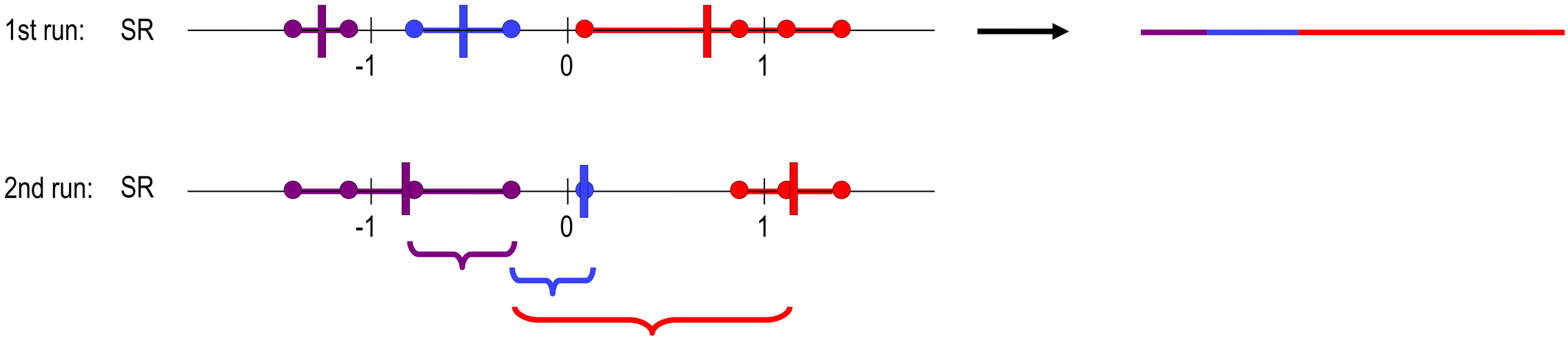
Total within-cluster variance:



k-means Clustering

1. Choosing the number of clusters: $k = 3$
2. Randomly assign cluster centers (centroids)
3. **Assign data points to the cluster with the nearest centroid**
4. **Update centroids to cluster mean**

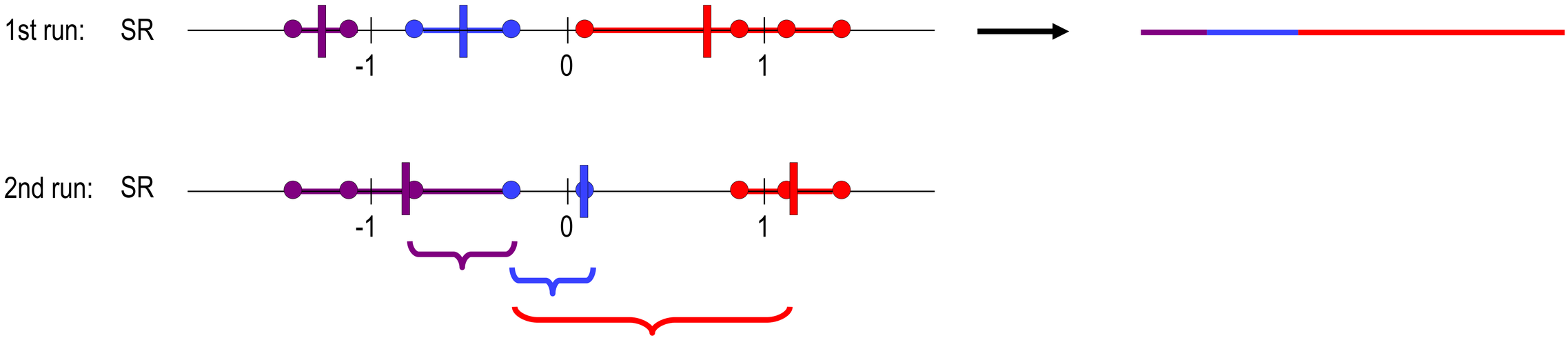
Total within-cluster variance:



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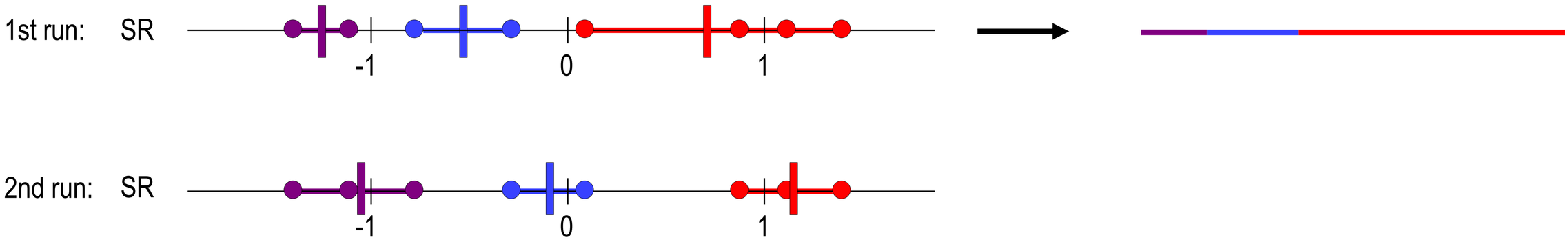
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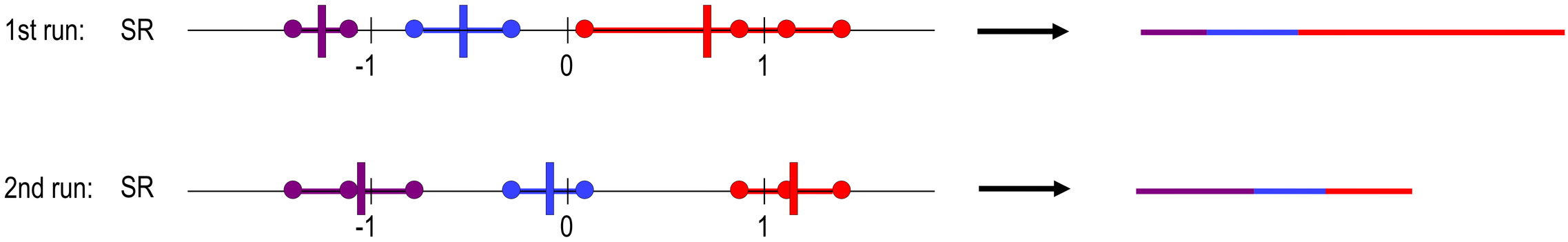
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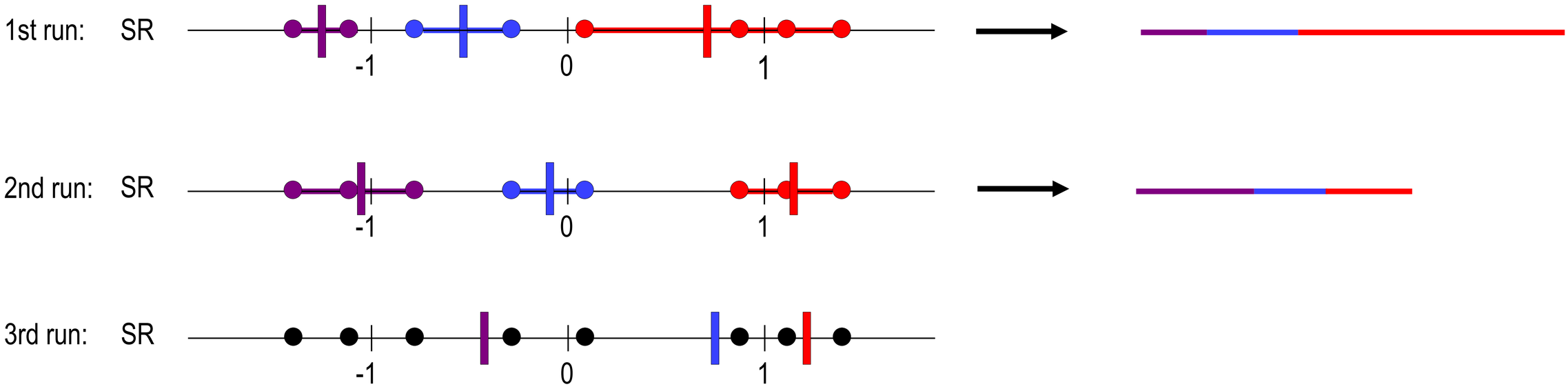
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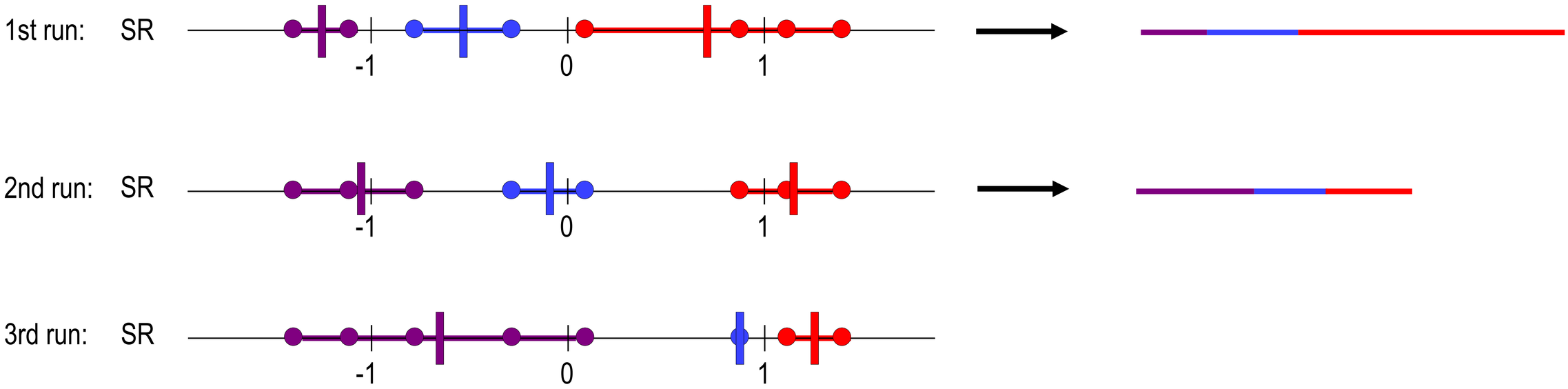
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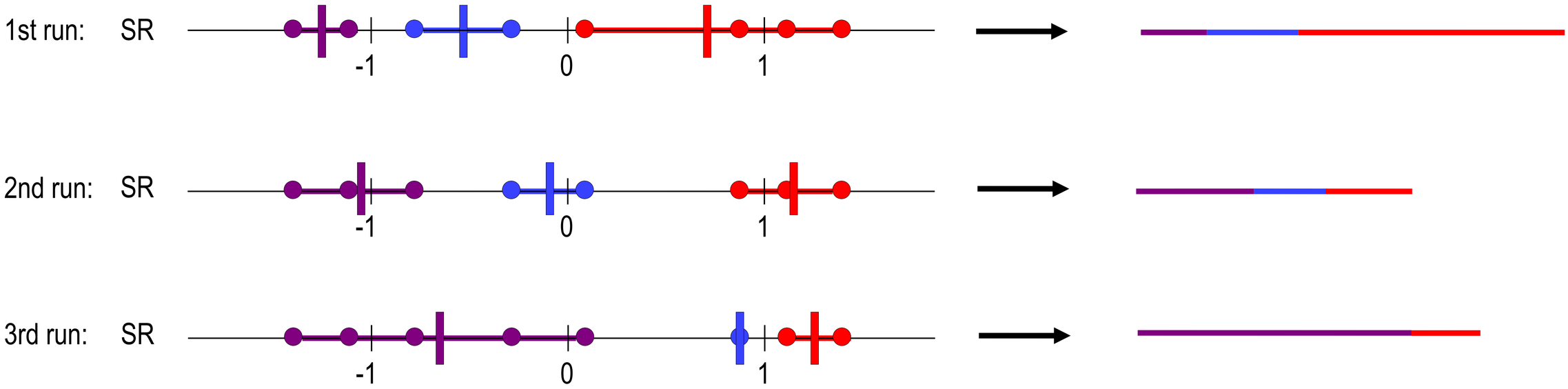
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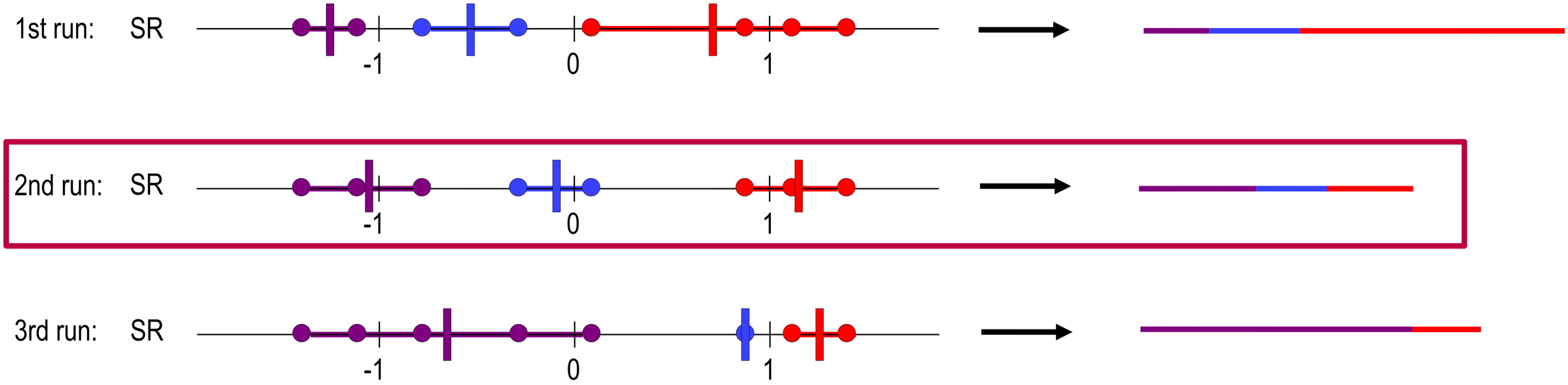


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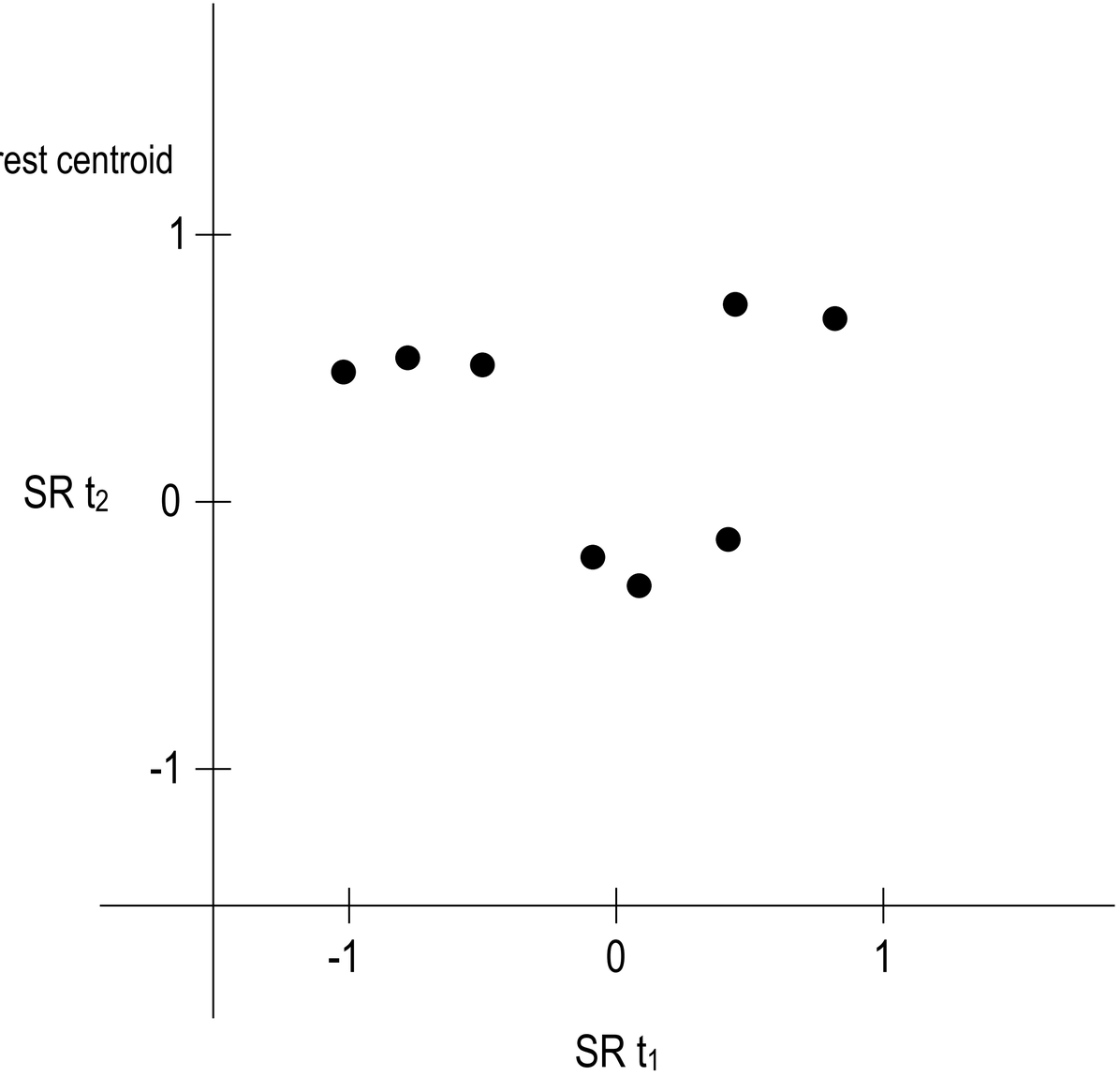
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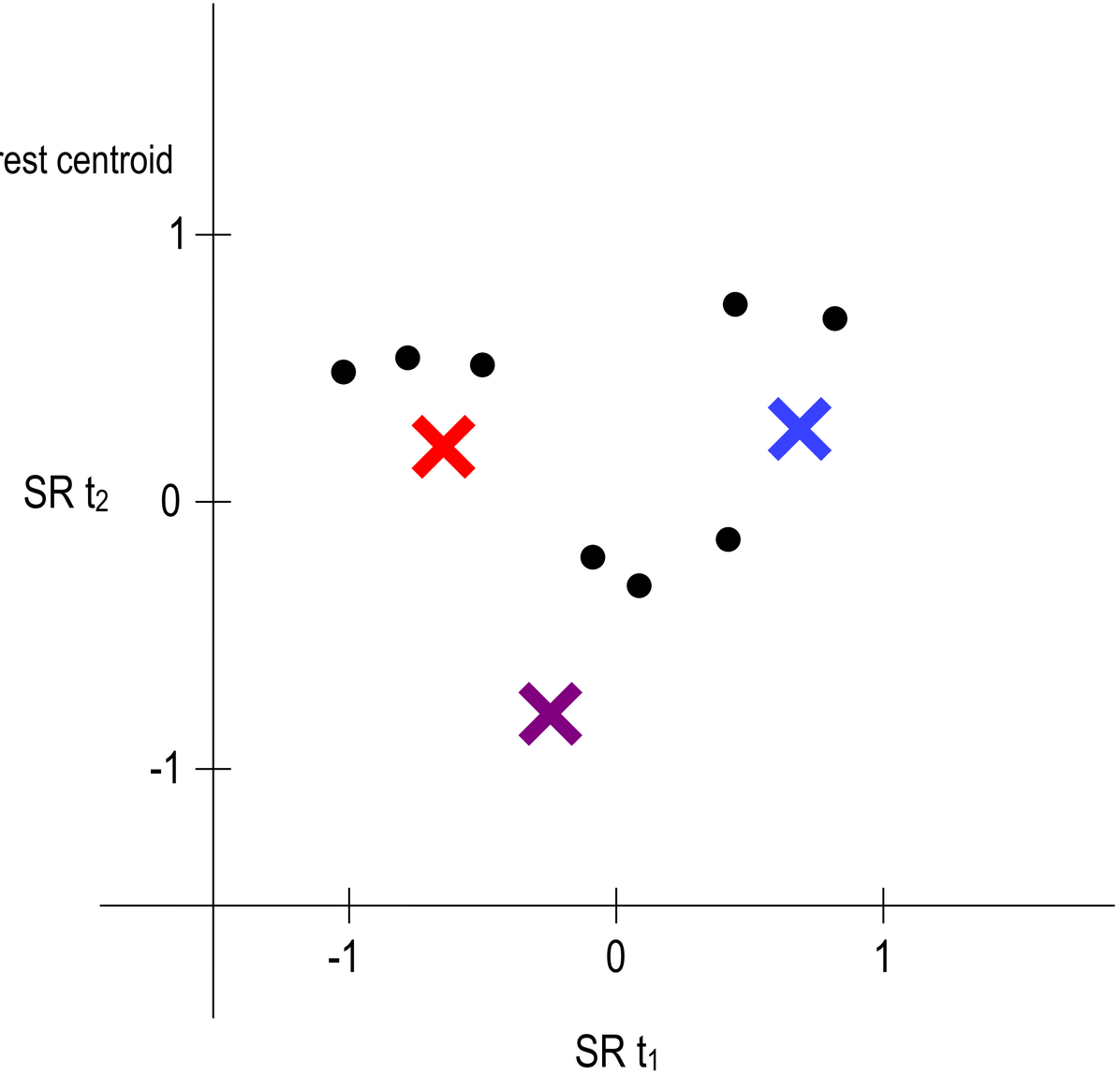
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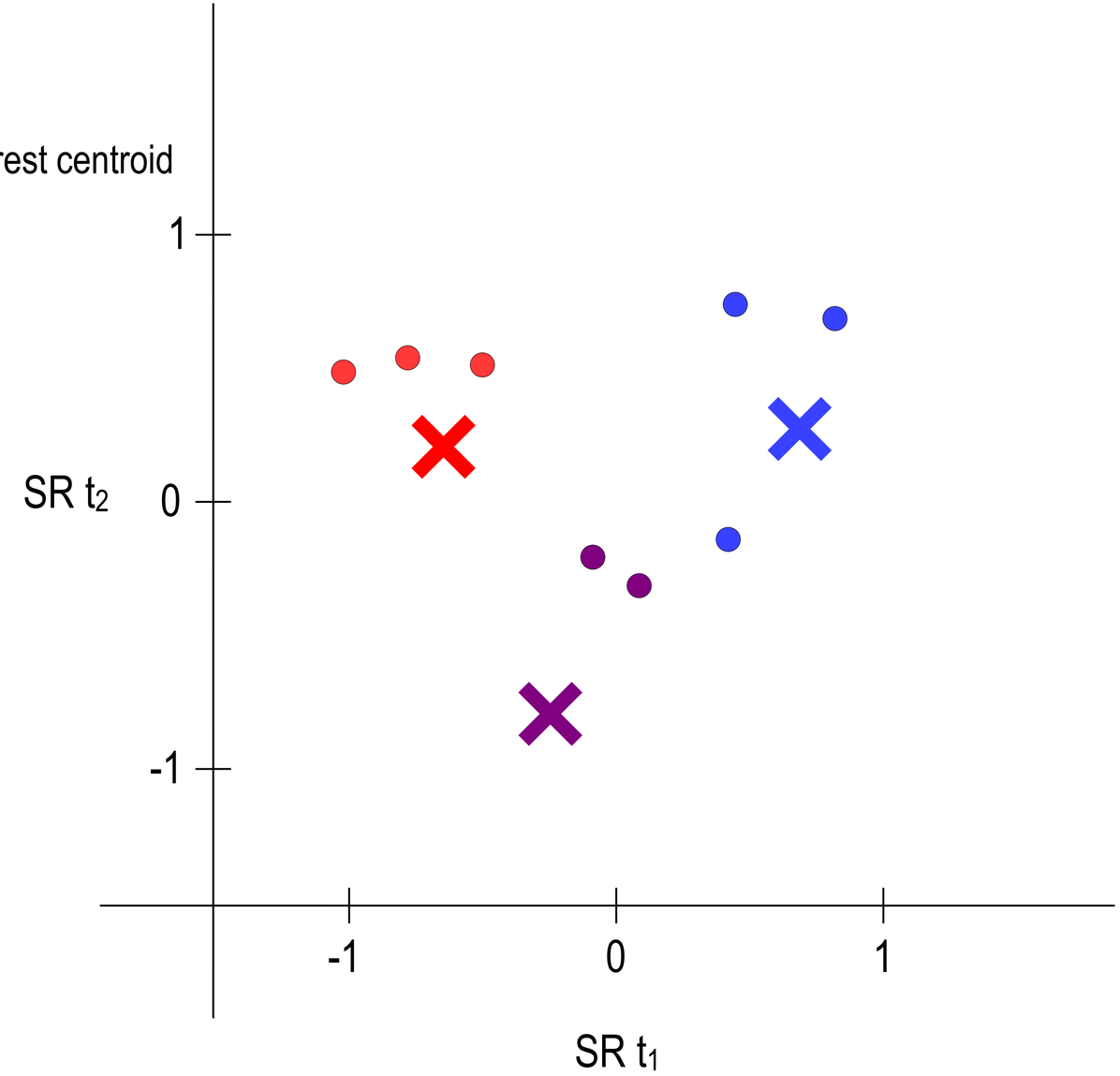
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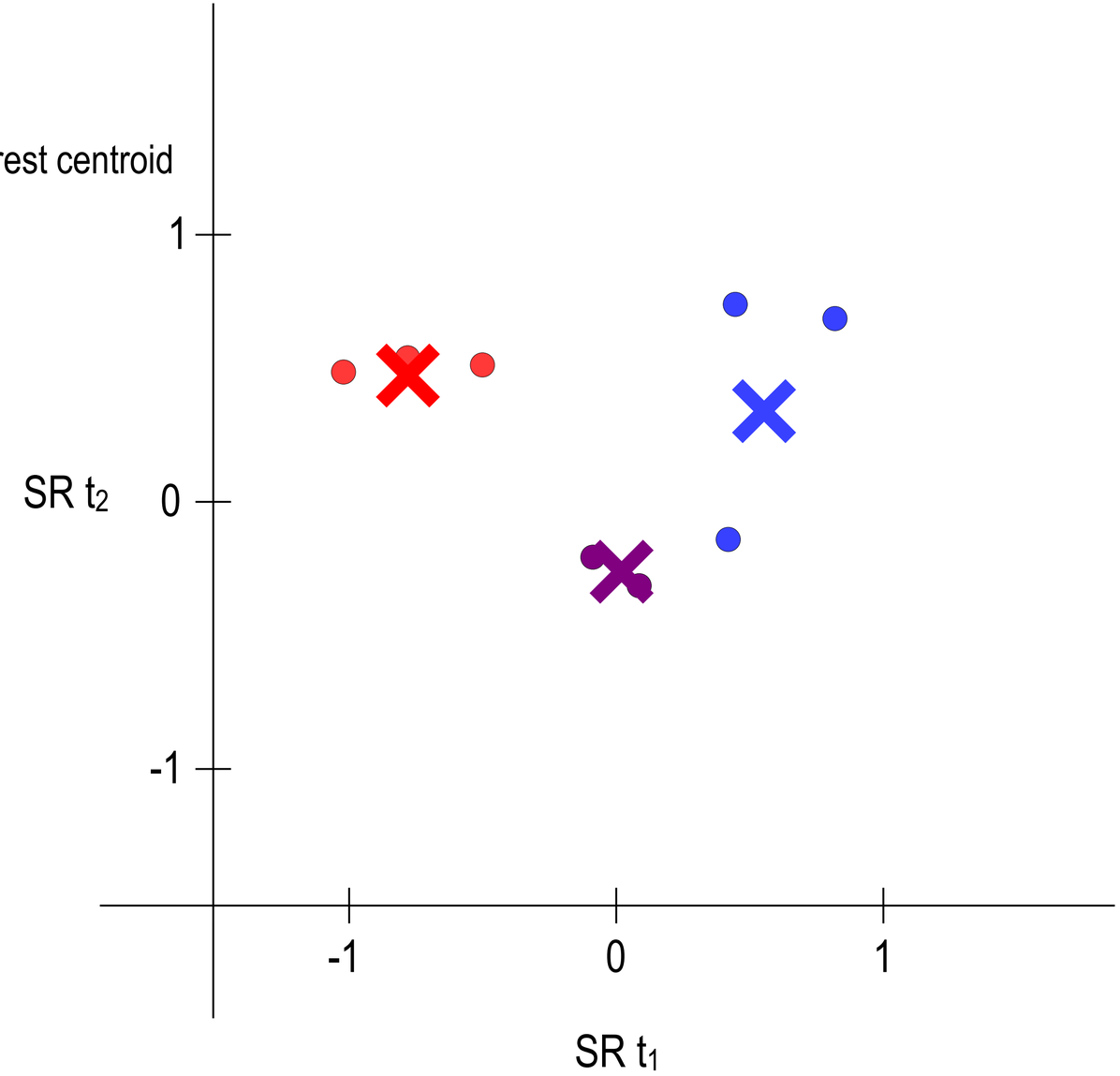
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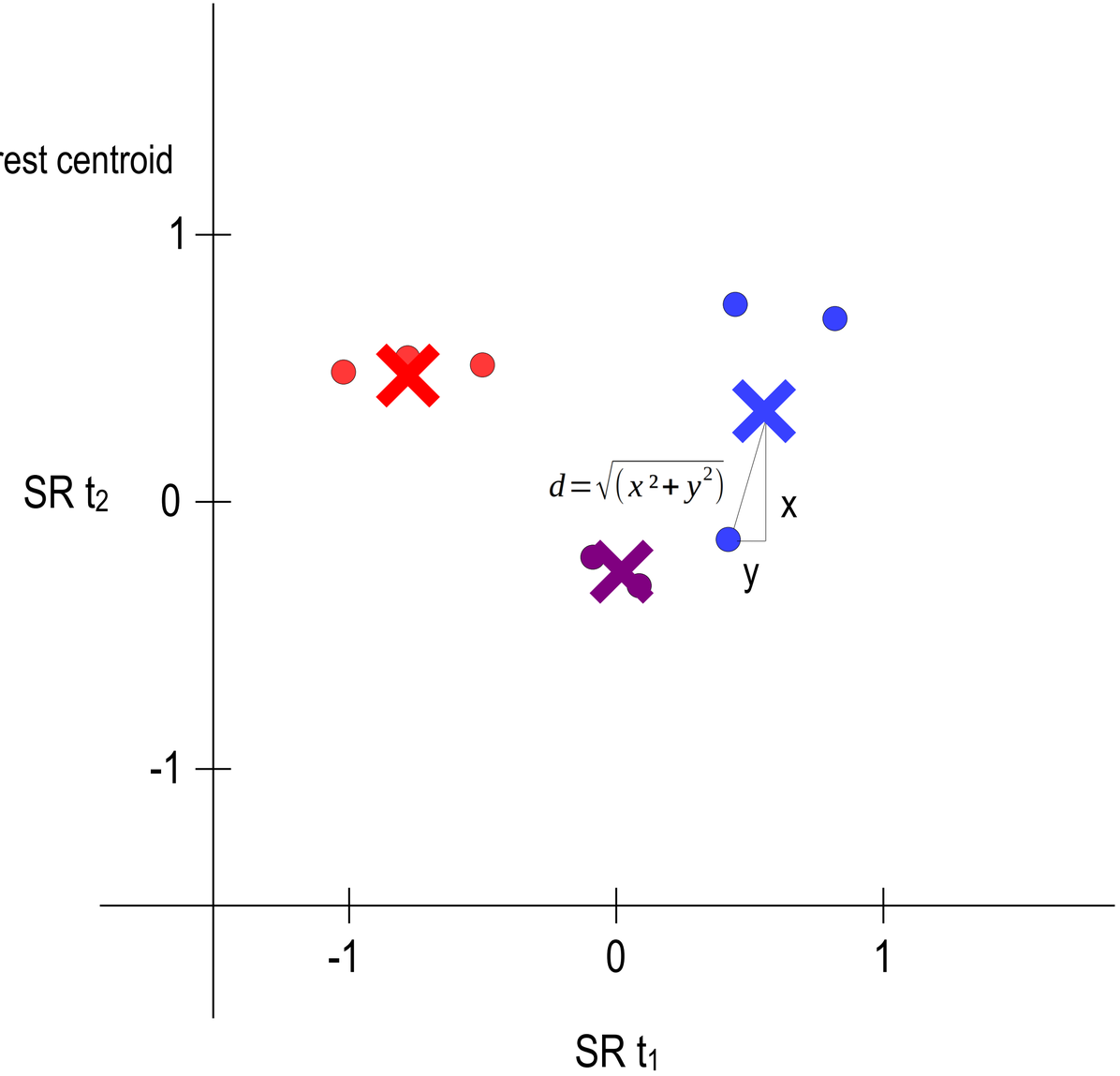
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